



Article: 263

**TOPICAL
WORKBOOK**

Volume-1 4024
MATHEMATICS
PAPER-1&2

- ✓ All Variants
- ✓ Mark Schemes Included
- ✓ Questions order new to old
- ✓ Writing space given



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Article No. 263

Mathematics⁴⁰²⁴

Paper-1 & 2 Topical Workbook with Mark Scheme
(Volume-1)
(2008-2018)

Features:

- ✓ All Variants
- ✓ Mark schemes included
- ✓ Questions order new to old
- ✓ Writing space given

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Sale Point: Shop No. 25-28 Lower Ground Floor, Haadia Haleema Centre, Ghazni Street, Urdu Bazar, Lahore

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Title	Mathematics O-Level P-1&2 Topical Workbook V-1 (Article# 263)
Published by	Read & Write Publications.
Printed by	Read & Write Publications.
Composing Department	Sharjeel Khan, Salman Buksh, Hayat, Shahab, Waqas, M. Awais, Usama Farooq
Graphics Department	Rashid Mehmood, Khalid Khursheed
Legal Advisor	Mian Tariq Ahmad (Advocate Supreme Court) Room No. 10, 11, 12 Al-Majeed Centre 1-Mozang Road, Lahore. Tel: 042-37236145, Fax: 042-37241367
Edition:	2019 Edition
Price	RS 800/-

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Note

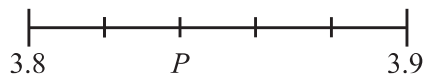
Complete syllabus of O-Level Mathematics Paper '1' and Paper '2' is divided in 4 volumes as given in the following table.

Volume 1	Volume 2	Volume 3	Volume 4
1. Number 2. Set Language and notation 3. Squares, square roots, cubes and cube roots 4. Directed numbers 5. Vulgar and decimal fractions and percentages 6. Ordering 7. Standard form 8. The four operations 9. Estimation 10. Limits of accuracy 11. Ratio, proportion, rate 12. Percentages 13. Use of an electronic calculator 14. Time 15. Money 16. Personal and small business finance 17. Algebraic representation and formulae 18. Algebraic manipulation 19. Indices	1. Solutions of equations and inequalities 2. Graphical representation of inequalities 3. Sequences 4. Variation 5. Graphs in practical situations 6. Graphs of functions 7. Function notation 8. Coordinate geometry 9. Geometrical terms 10. Geometrical constructions 11. Similarity and congruence	1. Symmetry 2. Angles 3. Loci 4. Measurers 5. Mensuration 6. Trigonometry	1. Vectors in two dimensions 2. Matrices 3. Transformations 4. Probability 5. Categorical, numerical and grouped data 6. Statistical diagrams 7. Everyday mathematics

CONTENTS

UNIT 1	Number (Paper-1) -----	3
	Answer Section -----	22
	Number (Paper-2) -----	24
	Answer Section -----	25
UNIT 2	Set Language And Notation (Paper-1) -----	26
	Answer Section -----	45
	Set Language And Notation (Paper-2) -----	47
	Answer Section -----	62
UNIT 3	Squares, Square Roots, Cubes And Cube Roots (Paper-1) -----	64
	Answer Section -----	67
	Squares, Square Roots, Cubes And Cube Roots (Paper-2) -----	68
	Answer Section -----	70
UNIT 4	Directed Numbers (Paper-1) -----	71
	Answer Section -----	73
UNIT 5	Vulgar And Decimal Fractions And Percentages (Paper-1) -----	74
	Answer Section -----	92
UNIT 6	Ordering (Paper-1) -----	94
	Answer Section -----	99
UNIT 7	Standard Form (Paper-1) -----	100
	Answer Section -----	118
	Standard Form (Paper-2) -----	120
	Answer Section -----	123
UNIT 8	The Four Operations (Paper-1) -----	124
	Answer Section -----	129
UNIT 9	Estimation (Paper-1) -----	130
	Answer Section -----	142
UNIT 10	Limits Of Accuracy (Paper-1) -----	144
	Answer Section -----	156
	Limits Of Accuracy (Paper-2) -----	157
	Answer Section -----	160
UNIT 11	Ratio Proportion, Rate (Paper-1) -----	161
	Answer Section -----	177
	Ratio Proportion, Rate (Paper-2) -----	179

	Answer Section -----	186
UNIT 12	Percentages (Paper-1) -----	187
	Answer Section -----	197
	Percentages (Paper-2) -----	198
	Answer Section -----	219
UNIT 13	Time (Paper-1) -----	221
	Answer Section -----	236
UNIT 14	Money (Paper-1) -----	237
	Answer Section -----	246
	Money (Paper-2) -----	247
	Answer Section -----	260
UNIT 15	Personal And Small Business Finance (Paper-1) -----	261
	Answer Section -----	263
	Personal And Small Business Finance (Paper-2) -----	264
	Answer Section -----	285
UNIT 16	Algebraic Representation And Formulae (Paper-1) -----	286
	Answer Section -----	296
	Algebraic Representation And Formulae (Paper-2) -----	297
	Answer Section -----	303
UNIT 17	Algebraic Manipulation (Paper-1) -----	304
	Answer Section -----	332
	Algebraic Manipulation (Paper-2) -----	335
	Answer Section -----	382
UNIT 18	Indices (Paper-1) -----	386
	Answer Section -----	402
	Indices (Paper-2) -----	404
	Answer Section -----	405

Unit 1: Number**Paper 1****1. O/N 17/P12/Q6/a****(a)**

The diagram shows a scale from 3.8 to 3.9, divided into five equal parts.

What is the value at the mark labelled P ?

Answer [1]

2. M/J 17/P12/Q1/b**(a)** Evaluate 0.2×0.006 .

Answer [1]

3. M/J 17/P11/Q1/b**(a)** Add one pair of brackets to make the statement below true.

$$2 \times 3 + 4 \times 5 = 70 \quad [1]$$

4. M/J 17/P11/Q14**(a)** Write 0.000 186 in standard form.

Answer [1]

(b) $s = 1.3 \times 10^7$ $t = 8 \times 10^8$

Giving each answer in standard form, find

(i) t^2 ,

Answer [1]

(ii) $t - s$.

Answer [2]

5. O/N 16/P12/Q1/a

(a) Evaluate $9.03 - (4.273 + 2.3)$.

Answer [1]

6. O/N 16/P11/Q2

The paper on a roll is 4.5 metres long.

Mary cuts as many pieces as possible, each of length 60 cm, from the roll.

(a) Calculate the number of pieces.

Answer [1]

(b) Calculate the length of paper that remains on the roll.

Answer cm [1]

7. M/J 16/P12/Q6

- (a) Express 96 as a product of its prime factors.

Answer [1]

- (b) 24 is a common factor of 96 and the integer n .
Given that n is less than 96, find the largest possible value of n .

Answer [1]

8. M/J 16/P11/Q1/a

- (a) Evaluate $12 - 6 \div 3 + 4$.

Answer [1]

9. M/J 16/P11/Q21

- (a) Express 500 as the product of its prime factors.

Answer [1]

(b) $M = 2 \times 3^2$ $N = 2^4 \times 3^2$

Find the values of p and q when

(i) $M \times N = 2^p \times 3^q$,

Answer $p = \dots\dots\dots q = \dots\dots\dots$ [1]

(ii) $M \div N = 2^p \times 3^q$,

Answer $p = \dots\dots\dots q = \dots\dots\dots$ [1]

(iii) $N^2 = 2^p \times 3^q$.

Answer $p = \dots\dots\dots q = \dots\dots\dots$ [1]

10. O/N 15/P12/Q1

(a) Evaluate 0.03×0.3 .

Answer..... [1]

(b) Evaluate $5 - 2(3 - 1.4)$.

Answer [1]

11. O/N 15/P11/Q1

(a) Work out $12 + 6 \div 3 + 1 \times 5$.

Answer [1]

(b) Work out $\frac{7}{9} - \frac{3}{5}$.

Answer [1]

12. O/N 15/P11/Q10/a,b

Here is a list of numbers.

−8 −5 −3 −2 0 2 4 9

(a) Write down two numbers from the list that have a difference of 10.

Answer and [1]

(b) Find the sum of the numbers in the list.

Answer [1]

13. M/J 15/P12/Q1

(a) Evaluate $\frac{1.3 + 2.9}{0.2}$.

Answer [1]

(b) Evaluate $2\frac{1}{4} \times \frac{1}{5}$.

Answer [1]

14. M/J 15/P11/Q1

(a) Evaluate $\frac{3}{8} + 1\frac{1}{3}$.

Answer [1]

(b) Evaluate $5 - 3(2 - 1.4)$.

Answer [1]

15. O/N 14/P11/Q1/a

(a) Write the number forty one thousand and six in figures.

Answer [1]

16. O/N 14/P11/Q2/b

- (a) Evaluate 0.4×0.2 .

Answer [1]

17. O/N 14/P11/Q4/b

x is an integer between 50 and 70 .

Write down the value of x when

- (a) x is a prime factor of 268.

Answer [1]

18. M/J 14/P12/Q1/b

- (a) Evaluate $0.018 \div 0.06$.

Answer [1]

19. M/J 14/P11/Q2

- (a) Evaluate $5 + 1 \times 0.3$.

Answer [1]

- (b) Evaluate $18 \div 0.2$.

Answer [1]

20. M/J 14/P11/Q8/b

- (a) Write $\frac{8}{25}$ as a decimal.

Answer [1]

21. M/J 14/P11/Q15

- (a) Find an integer r such that $r > 5$ and $5r - 1$ is a square number.

Answer $r =$ [1]

- (b) Find the value of s which makes $8s + 2$ a prime number.

Answer $s =$ [1]

- (c) Write down an irrational number between 7 and 8.

Answer [1]

22. O/N 13/P12/Q2

- (a) Evaluate $3\frac{1}{4} - 1\frac{4}{5}$.

Answer [1]

(b) Evaluate 3.01×0.02 .

Answer [1]

23. O/N 13/P11/Q1/a

(a) Evaluate $2\frac{3}{4} - 1\frac{13}{16}$.

Answer [1]

24. O/N 13/P11/Q2/a

(a) Evaluate 0.02×1.2 .

Answer [1]

25. M/J 13/P12/Q3/a

(a) Write these lengths in order of size, starting with the shortest.

500m

5cm

50km

500mm

Answer
shortest

[1]

26. M/J 13/P12/Q8/b,c

- (a) Omar thinks of a **two-digit** number.
It is a factor of 78.
It is a prime number.
What is his number?

Answer [1]

- (b) Write down an irrational number between 1 and 2.

Answer [1]

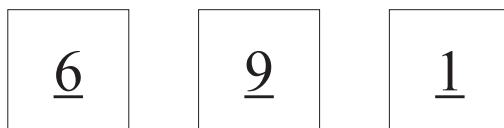
27. M/J 13/P12/Q9/b,c

- (a) $\sqrt{131}$ lies between two consecutive integers.
Complete the inequality below with these integers.

Answer $< \sqrt{131} < \dots\dots\dots$ [1]

- (b) Add brackets to the statement below to make it correct.

$$3 \times 2 + 1^2 = 49 \quad [1]$$

28. M/J 13/P11/Q12/a,b

The three cards above can be rearranged to make three-digit numbers, for example 916.
Arrange the three cards to make

- (a) the three-digit number that is closest to 650,

Answer [1]

- (b) the three-digit number that is a multiple of 7,

Answer [1]

29. O/N 12/P12/Q1/a

- (a) Evaluate $8 + 2 \times 1.3$.

Answer.....[1]

30. O/N 12/P12/Q10

- (a) Express 180 as the product of its prime factors.

Answer[1]

- (b) $\sqrt{180}$ can be expressed in the form $p\sqrt{q}$, where p and q are integers.

Find the smallest value of $p + q$.

Answer[1]

31. O/N 12/P11/Q2/a

- (a) Evaluate $0.7 + 0.2 \times 0.3$.

Answer[1]

32. M/J 12/P11/Q9

Buses following route A leave the bus station every five minutes.

Buses following route B leave the bus station every six minutes.

Buses following route C leave the bus station every nine minutes.

Three buses, following routes A, B and C, leave together at 13 00.

What is the next time when buses following all three routes leave the bus station together?

Answer [2]

33. M/J 12/P11/Q12

0.2	2	$\sqrt{2}$	$\frac{1}{3}$	0.83	8	81
-----	---	------------	---------------	------	---	----

From the numbers listed above, write down

(a) a prime number,

Answer [1]

(b) a cube number,

Answer [1]

(c) an irrational number.

Answer [1]

34. O/N 11/P12/Q1

(a) Evaluate $2\frac{3}{4} - 1\frac{7}{9}$.

Answer [1]

(b) Evaluate $0.7 - 0.1 \times 3$.

Answer [1]

35. M/J 11/P12/Q14

(a) Express 108 as a product of its prime factors.

Answer [1]

(b) Written as products of their prime factors, $N = 2^p \times 5^q \times 7^r$ and $500 = 2^2 \times 5^3$.

The highest common factor of N and 500 is $2^2 \times 5^2$.

The lowest common multiple of N and 500 is $2^3 \times 5^3 \times 7$.

Find p , q and r .

Answer $p = \dots$, $q = \dots$, $r = \dots$ [2]

36. M/J 11/P11/Q2/b

- (a) Write down an irrational number between 3 and 4.

Answer [1]

37. M/J 11/P11/Q4

Write down

- (a) a square number that is a factor of 75,

Answer [1]

- (b) a cube number that is a multiple of 24.

Answer [1]

38. O/N 10/P12/Q1, O/N 10/P13/Q1

- (a) Evaluate $3\frac{1}{7} - 2\frac{1}{3}$.

Answer (a) [1]

(b) Evaluate $\frac{2}{9} \times 1\frac{7}{8}$, giving your answer as a fraction in its lowest terms.

Answer (b)[1]

40. M/J 10/P12/Q1, M/J 10/P13/Q1

Evaluate

(a) $1.5 - 0.2 \times 4$,

Answer (a)[1]

(b) $4.2 \div 0.07$.

Answer (b)[1]

41. M/J 10/P12/Q9, M/J 10/P13/Q9

Written as a product of prime factors, $168 = 2^3 \times 3 \times 7$.

(a) Express 140 as a product of its prime factors.

Answer (a)[1]

- (b) Find the highest common factor of 168 and 140.

Answer (b)[1]

- (c) Find the smallest positive integer, n , such that $168n$ is a square number.

Answer (c)[1]

42. M/J 10/P11/Q2

- (a) Evaluate $10 - 8 \div 2 + 3$.

Answer (a)[1]

- (b) Find 20% of 60 cm.

Answer (b)cm [1]

43. O/N 09/P01/Q4

The numbers 294 and 784, written as the product of their prime factors, are

$$294 = 2 \times 3 \times 7^2, \quad 784 = 2^4 \times 7^2.$$

Find

- (a) the largest integer which is a factor of both 294 and 784,

Answer (a)[1]

(b) $\sqrt{784}$.

Answer (b)[1]

44. M/J 09/P01/Q1/a

(a) Evaluate $17 - 5 \times 3 + 1$.

Answer (a)[1]

45. M/J 09/P01/Q3/b

(a) Write down the two prime numbers between 30 and 40.

Answer (b) [1]

46. M/J 09/P01/Q5/a

(a) Evaluate 0.5×0.007 .

Answer (a)[1]

47. M/J 09/P01/Q6

- (a) Write down all the factors of 18.

Answer (a) [1]

- (b) Write 392 as the product of its prime factors.

Answer (b) [1]

48. M/J 09/P01/Q17/b

- (a) Given that y is an integer and $-3 < 2y - 6 < 4$, list the possible values of y .

Answer (b) [2]

49. O/N 08/P01/Q1

Evaluate

(a) 0.3×0.06 ,

(b) $0.4 + 0.3 \times 5$.

Answer (a) [1]

(b) [1]

50. M/J 08/P01/Q1

Evaluate

(a) $\frac{1}{2} - \frac{3}{7}$,

(b) $2\frac{2}{3} \times 1\frac{3}{4}$.

Answer (a)[1]

(b)[1]

51. M/J 08/P01/Q2

Evaluate

(a) $25 - 18.3$,

(b) 1.7×0.03 .

Answer (a)[1]

(b)[1]

52. M/J 08/P01/Q5It is given that $68.2 \times 0.235 = 16.027$.

Hence evaluate

(a) 0.0682×2350 ,

(b) $160.27 \div 0.0235$.

Answer (a)[1]

(b)[1]

Answers Section**Paper 1**

- | | | | |
|-------------------------------------|----|--|---|
| 1. O/N 17/P12/Q6/a | | 13. M/J 15/P12/Q1 | |
| (a) 3.84 | 1 | (a) 21 | 1 |
| 2. M/J 17/P12/Q1/b | | (b) $\frac{9}{20}$ oe | 1 |
| (a) 0.0012 oe | 1 | 14. M/J 15/P11/Q1 | |
| 3. M/J 17/P11/Q1/b | | (a) $1\frac{17}{24}, \frac{41}{24}$ oe | 1 |
| (a) $2 \times (3 + 4) \times 5$ cao | 1 | (b) 3.2 oe | 1 |
| 4. M/J 17/P11/Q14 | | 15. O/N 14/P11/Q1/a | |
| (a) 1.86×10^{-4} | 1 | (a) 41 006 | 1 |
| (b) (i) 6.4×10^{17} | 1 | 16. O/N 14/P11/Q2/b | |
| (ii) 7.87×10^8 | 2 | (a) (0).08 | 1 |
| 5. O/N 16/P12/Q1/a | | 17. O/N 14/P11/Q4/b | |
| (a) 2.457 | 1 | (a) 67 | 1 |
| 6. O/N 16/P11/Q2 | | 18. M/J 14/P12/Q1/b | |
| (a) 7 | 1 | (a) 0.3oe | 1 |
| (b) 30 | 1 | 19. M/J 14/P11/Q2 | |
| 7. M/J 16/P12/Q6 | | (a) 5.3 | 1 |
| (a) $2^5 \times 3$ | 1* | (b) 90 | 1 |
| (b) 72 | 1 | 20. M/J 14/P11/Q8/b | |
| 8. M/J 16/P11/Q1/a | | (a) (0).32 | 1 |
| (a) 14 | 1 | 21. M/J 14/P11/Q15 | |
| 9. M/J 16/P11/Q21 | | (a) 10 etc. | 1 |
| (a) $2^2 \times 5^3$ | 1 | (b) 0 | 1 |
| (b) (i) $p = 5$ and $q = 4$ | 1 | (c) $\sqrt{50}$ etc. | 1 |
| (ii) $p = -3$ and $q = 0$ | 1 | 22. O/N 13/P12/Q2 | |
| (iii) $p = 8$ and $q = 4$ | 1 | (a) $1\frac{9}{20}$ | |
| 10. O/N 15/P12/Q1 | | (b) 0.0602 | |
| (a) 0.009(0...) | 1 | 23. O/N 13/P11/Q1/a | |
| (b) 1.8 | 1 | (a) $\frac{15}{16}$ oe | 1 |
| 11. O/N 15/P11/Q1 | | 24. O/N 13/P11/Q2/a | |
| (a) 19 | 1 | (a) 0.024 | 1 |
| (b) $\frac{8}{45}$ oe | 1 | | |
| 12. O/N 15/P11/Q10/a,b | | | |
| (a) -8 and 2 | 1 | | |
| (b) -3 | 1 | | |

- 25. M/J 13/P12/Q3/a**
(a) 5 cm, 500 mm, 500 m, 50 km 1
- 26. M/J 13/P12/Q8/b,c**
(a) 13 1
(b) Any irrational number in range $1 < n < 2$ 1
- 27. M/J 13/P12/Q9/b,c**
(a) $11 (< \sqrt{131} <) 12$ 1
(b) $(3 \times 2 + 1)^2 = 49$ 1
- 28. M/J 13/P11/Q12/a,b**
(a) 619 1
(b) 196 1
- 29. O/N 12/P12/Q1/a**
(a) 10.6 1
- 30. O/N 12/P12/Q10**
(a) $2^2 \times 3^2 \times 5$ oe 1
(b) 11 www 1
- 31. O/N 12/P11/Q2/a**
(a) 0.76 oe 1
- 32. M/J 12/P11/Q9**
14 30 2
- 33. M/J 12/P11/Q12**
(a) 2 1
(b) 8 1
(c) $\sqrt{2}$ 1
- 34. O/N 11/P12/Q1**
(a) $\frac{35}{36}$ 1
(b) 0.4 1
- 35. M/J 11/P12/Q14**
(a) $2^2 \times 3^3$ 1
(b) $(p=) 3, (q=) 2, (r=) 1$ 2
- 36. M/J 11/P11/Q2/b**
(a) $\pi, \sqrt{10}, \dots$ 1
- 37. M/J 11/P11/Q4**
(a) 1 or 25 1
(b) 216 1
- 38. O/N 10/P12/Q1, O/N 10/P13/Q1**
(a) $\frac{17}{21}$ oe 1
(b) $\frac{5}{12}$ cao 1
- 39. O/N 10/P12/Q2, O/N 10/P13/Q2**
(a) 70 1 (b) 4.05 1
- 40. M/J 10/P12/Q1, M/J 10/P13/Q1**
(a) 0.7 1 (b) 60 1
- 41. M/J 10/P12/Q9, M/J 10/P13/Q9**
(a) $2^2 \times 5 \times 7$ 1
(b) 28 1
(c) 42 1
- 42. M/J 10/P11/Q2**
(a) 9 1 (b) 12 1
- 43. O/N 09/P01/Q4**
(a) $98, 2 \times 7^2, 2 \times 7 \times 7$ 1
(b) 28 1
- 44. M/J 09/P01/Q1/a**
(a) 3 1
- 45. M/J 09/P01/Q3/b**
(a) 31, 37 1
- 46. M/J 09/P01/Q5/a**
(a) (0).0035
- 47. M/J 09/P01/Q6**
(a) 1,2,3,6,9,18 1
(b) $2^3 \times 7^2$ 1
- 48. M/J 09/P01/Q17/b**
(a) 2,3,4 2
- 49. O/N 08/P01/Q1**
(a) 0.018 or equiv. 1
(b) 1.9 or equiv. 1
- 50. M/J 08/P01/Q1**
(a) $\frac{1}{14}$ 1 (b) $4\frac{2k}{3k}$ 1
- 51. M/J 08/P01/Q2**
(a) 6.7 oe 1 (b) (0).051 oe 1
- 52. M/J 08/P01/Q5**
(a) 160.27 1 (b) 6820 1

Paper 2

1. M/J 18/P22/Q4/b

- (a) Find the lowest common multiple (LCM) of 140 and 770.

Answer [2]

2. M/J 18/P21/Q1/c(i)

- (a) (i) Write 540 as the product of its prime factors.

Answer [2]

Answers Section**Paper 2**

1. M/J 18/P22/Q4/ b

(a) 1540 2

2. M/J 18/P21/Q1/ c(i)

(a) (i) $2^2 \times 3^3 \times 5$ 2

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